

EXP.NO: 5	Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.
DATE:13/02/2025	

AIM:

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

ALGORITHM:

1. Input:

- The user fills out a web form with their **name** and **email**.
- The form will use both **GET** and **POST** methods for data submission.

2. Steps for Creating and Handling the HTTP GET and POST Requests:

1. Create an HTML Form:

- Design an HTML form with two input fields: one for the **name** and one for the **email**.
- Create two buttons to allow the user to submit the form using **GET** or **POST** methods.
- Use the action attribute to define the **Servlet URL** and specify the form method (either GET or POST).

2. Create the Servlet to Handle GET and POST Requests:

- Write a Java Servlet class that will handle **both GET and POST requests**.
- The servlet will have two methods:
 - doGet() to handle the GET requests.
 - doPost() to handle the POST requests.
- In both methods, retrieve the form data using request.getParameter().
- In the doGet() method, process and display the data in the URL.
- In the doPost() method, process and display the data in the HTTP request body.

3. Handle GET Requests in the Servlet:

- In the `doGet()` method, use `request.getParameter()` to retrieve the name and email submitted via GET.
- Output the data as a response, including the **name** and **email** in the URL (visible to the user).

4. Handle POST Requests in the Servlet:

- In the `doPost()` method, use `request.getParameter()` to retrieve the name and email submitted via POST.
- Output the data as a response, displaying the **name** and **email** without exposing them in the URL (more secure).

5. Return the Output:

- The Servlet should return an HTML page displaying the **name** and **email** entered by the user.
- For the **GET method**, the output should be visible in the URL.
- For the **POST method**, the output should be visible in the response body, not the URL.

6. Deploy and Test the Servlet:

- Compile the Servlet class and deploy it to a **Servlet container** (like **Apache Tomcat**).
- Test the form using **GET** and **POST** methods to see the difference in data submission.

7. Observe and Compare GET and POST Results:

- Test the **GET method**: The form data is sent in the URL.
- Test the **POST method**: The form data is sent in the HTTP request body, not visible in the URL.

Code:

userform.java

```
package com.example;
```

```
import java.io.IOException;
```

```
import java.io.PrintWriter;

import javax.servlet.ServletException;

import javax.servlet.annotation.WebServlet;

import javax.servlet.http.HttpServlet;

import javax.servlet.http.HttpServletRequest;

import javax.servlet.http.HttpServletResponse;


@WebServlet("/userform")

public class userform extends HttpServlet {

    private static final long serialVersionUID = 1L;


    // common method for GET & POST

    protected void processRequest(HttpServletRequest request, HttpServletResponse
response, String method)

        throws ServletException, IOException {


        String name = request.getParameter("name");

        String slot = request.getParameter("slot");

        String hours = request.getParameter("hours");


        response.setContentType("text/html");

        PrintWriter out = response.getWriter();


        out.println("<html><head><title>Parking Booking Confirmation</title>");

        out.println("<style>");

        out.println("body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
background-color: #121212; color: #f0f0f0; display:flex; justify-content:center; align-
items:center; height:100vh; margin:0; }");
```

```

        out.println(".container { background-color:#1f1f1f; padding:30px; border-radius:12px;
box-shadow:0 0 15px rgba(0,0,0,0.5); text-align:center; width:400px; }");

        out.println("h1 { color:#3399ff; margin-bottom:20px; }");

        out.println("p { font-size:18px; margin:10px 0; }");

        out.println("a { display:inline-block; margin-top:20px; text-decoration:none;
color:#121212; background-color:#3399ff; padding:10px 20px; border-radius:8px; font-
weight:bold; transition:0.3s; }");

        out.println("a:hover { background-color:#2673cc; }");

        out.println("</style></head><body>");

        out.println("<div class='container'>");

        out.println("<h1>Booking via " + method + "</h1>");

        out.println("<p><strong>Name:</strong> " + (name != null ? name : "Not Provided") +
"</p>");

        out.println("<p><strong>Slot Number:</strong> " + (slot != null ? slot : "Not Provided")
+ "</p>");

        out.println("<p><strong>Hours:</strong> " + (hours != null ? hours : "Not Provided") +
"</p>");

        out.println("<a href='index.html'>Back to Booking Form</a>");

        out.println("</div></body></html>");

    }

```

```

protected void doGet(HttpServletRequest request, HttpServletResponse response)

```

```

    throws ServletException, IOException {

        processRequest(request, response, "GET");

    }

```

```

protected void doPost(HttpServletRequest request, HttpServletResponse response)

```

```

    throws ServletException, IOException {

        processRequest(request, response, "POST");

    }

```

```
}
```

index.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Movie Ticket Booking - GET vs POST</title>
```

```
<style>
```

```
  body {
```

```
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
```

```
    background-color: #121212;
```

```
    color: #f0f0f0;
```

```
    display: flex;
```

```
    justify-content: center;
```

```
    align-items: center;
```

```
    height: 100vh;
```

```
    margin: 0;
```

```
  }
```

```
  form {
```

```
    background-color: #1f1f1f;
```

```
    padding: 30px;
```

```
    border-radius: 12px;
```

```
    box-shadow: 0 0 15px rgba(0,0,0,0.5);
```

```
    width: 400px;
```

```
  }
```

```
  h1 {
```

```
text-align: center;  
color: #3399ff;  
margin-bottom: 20px;  
}
```

```
label {  
    display: block;  
    margin-top: 15px;  
    font-weight: bold;  
    color: #ddd;  
}
```

```
input[type="text"], input[type="number"] {  
    width: 100%;  
    padding: 10px;  
    margin-top: 5px;  
    border-radius: 8px;  
    border: 1px solid #444;  
    background-color: #2a2a2a;  
    color: #f0f0f0;  
}
```

```
input[type="submit"] {  
    margin-top: 20px;  
    width: 48%;  
    padding: 12px;  
    border: none;  
    border-radius: 8px;  
    background-color: #3399ff;  
    color: #121212;  
    font-weight: bold;
```

```
    font-size: 16px;

    cursor: pointer;

    transition: 0.3s;

}

input[type="submit"]:hover {

    background-color: #2673cc;

}

.buttons {

    display: flex;

    justify-content: space-between;

}

</style>

</head>

<body>

<form action="userform" method="post">

    <h1>Movie Ticket Booking</h1>

    <label for="name">Your Name:</label>

    <input type="text" name="name" id="name" required>

    <label for="slot">Number of Event:</label>

    <input type="number" name="slot" id="slot" min="1" max="50" required>

    <label for="hours">Number of Tickets:</label>

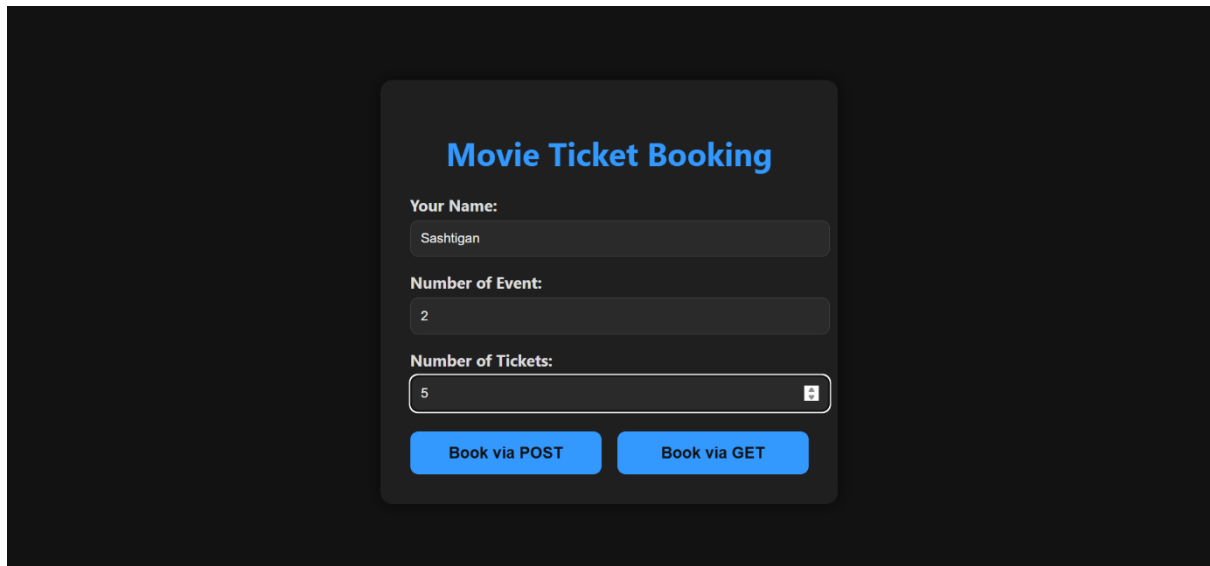
    <input type="number" name="hours" id="hours" min="1" required>

    <div class="buttons">

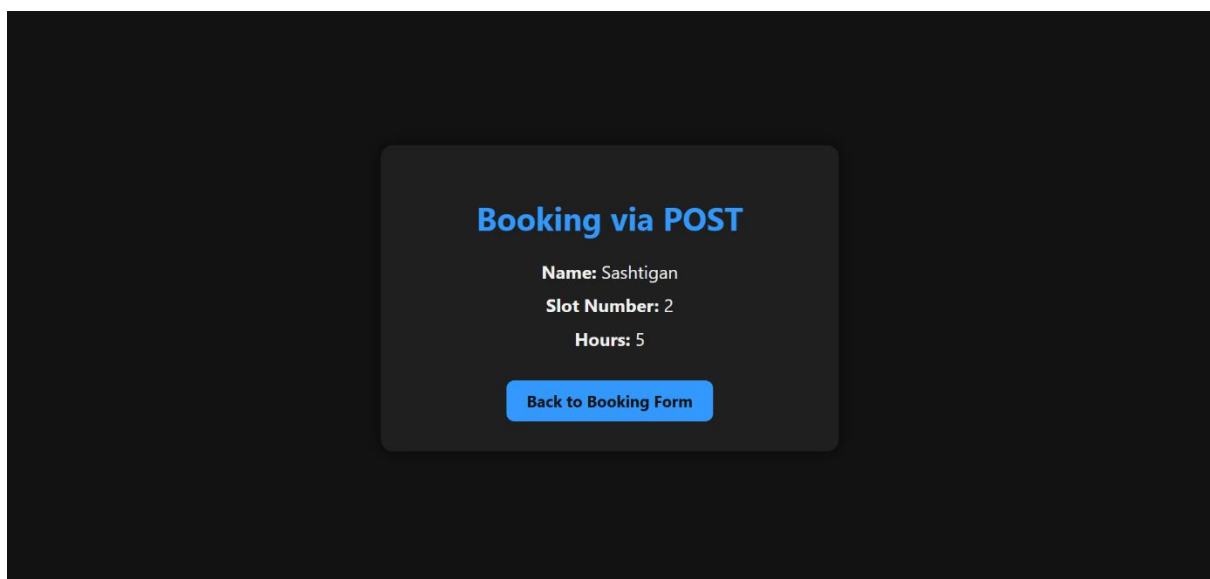
        <input type="submit" value="Book via POST">
```

```
<input type="submit" formaction="userform" formmethod="get" value="Book via  
GET">  
  
</div>  
  
</form>  
  
</body>  
  
</html>
```

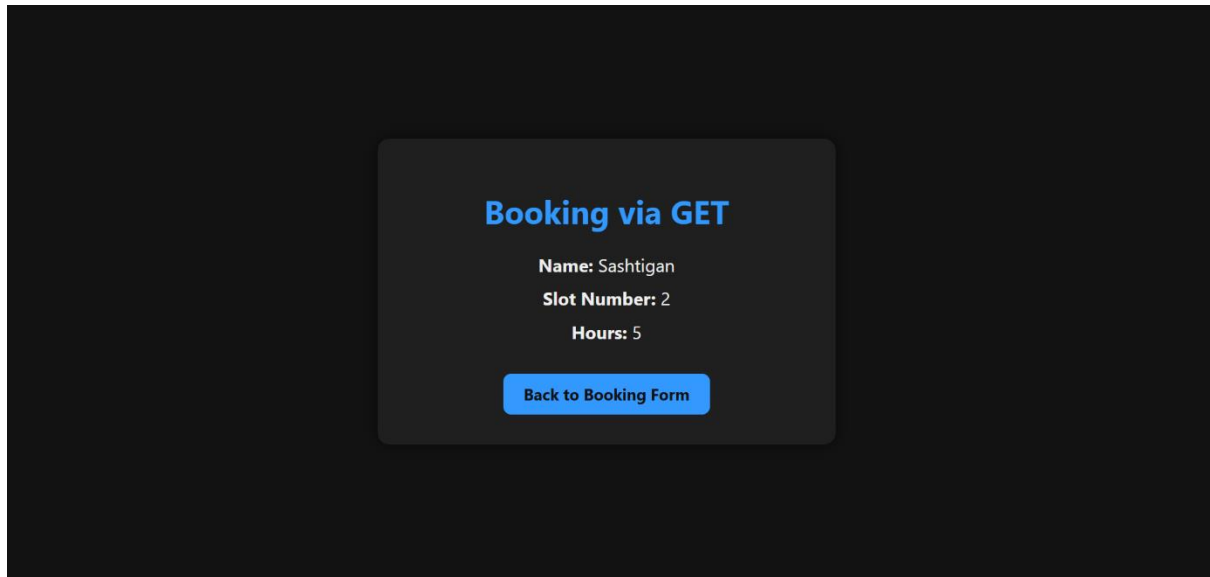
Output:



The screenshot shows a web form titled "Movie Ticket Booking" on a dark background. The form contains three input fields: "Your Name:" with the value "Sashtigan", "Number of Event:" with the value "2", and "Number of Tickets:" with the value "5". Below the inputs are two blue buttons: "Book via POST" and "Book via GET".



The screenshot shows a confirmation page titled "Booking via POST" on a dark background. It displays the booking details: "Name: Sashtigan", "Slot Number: 2", and "Hours: 5". At the bottom is a blue button labeled "Back to Booking Form".



RESULT:

A Servlet is created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.